



Databases and Applications

Unit III

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Computer Applications

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MongoDB Query Filters and Operators

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Computer Applications



► Comparison Operators

- \$eq – equals
- \$gt, \$lt – greater/less than
- \$gte, \$lte – greater/less than or equal

► Examples

```
db.students.find( { age: { $gt: 20 } } )
```

```
db.students.find( { dept: { $eq: "CSE" } } )
```



► Frequent beginner errors

- Writing `age > 20` instead of using `$gt`.
- Missing braces: must use `{ age: { $gt: 20 } }`.
- Forgetting that string comparison matches exactly.

► Correct pattern

```
db.students.find({ age: { $lte: 22 } })
```



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MongoDB Logical Operators

► Logical Operators

- \$and – combine multiple conditions
- \$or – match at least one condition
- \$not, \$nor – negation patterns

► Examples

```
db.students.find({ $and:  [{dept:"CSE"}, {age:{  
$gt:20 }}] })
```

```
db.students.find({ $or:  [{age:21}, {dept:"ECE"}]  
})
```



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Logical Operators – Usage Guide

► When to use

- Use \$and when filtering on 2+ conditions.
- Use \$or to search alternatives.
- Use \$not only inside another condition.

► Common mistakes

- Writing {age:21, \$or:[...]} (where ordering fails).
- Forgetting arrays inside \$and and \$or.
- Using SQL syntax: dept="CSE" AND age>20.



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MongoDB Array Operators

► Array Operators

- \$in - match any value from a list
- \$all - must contain all listed values
- \$size - array length equals

► Examples

```
db.students.find({ skills: { $in: ["Python","SQL"]  
} })
```

```
db.students.find({ skills: {  
$all: ["Python","MongoDB"] } })
```



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Array Queries – Mistakes to Avoid

► Common mistakes

- Expecting \$in to behave like AND.
- Using "skills.python" instead of array filters.
- Misusing \$size expecting ranges.

► Correct usage

```
db.students.find({ skills: { $size: 2 } })
```

Use \$all for “must contain both”.



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Element Operators – exists, type

► Element Operators

- \$exists – checks if a field is present.
- \$type – matches data type.

► Examples

```
db.students.find({ address: { $exists:true } })  
db.students.find({ age: { $type:"int" } })
```



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Querying Nested Fields

► Using dot notation

```
db.students.find({ "address.city": "Bengaluru" })
```

► When to use

- For structured data such as addresses, contacts, marks.
- Easier than joins in SQL.

► Common mistakes

- Using single quotes inside shell.
- Typing `address.city = "Bengaluru"` (SQL syntax).



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CELEBRATING 50 YEARS

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